

IIT JAM Mathematics (MA) 2026

Supplemental Cheat Sheet

Covering missing syllabus topics: 1D Max/Min, L'Hospital's Rule, Power Series Ops, & Total Derivative

1. Functions of One Real Variable

1.1 L'Hospital's Rule

Used to evaluate limits of indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , ∞^0 .

- **Standard Form:** Suppose $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$ (or $\pm\infty$). If f and g are differentiable near c and $g'(x) \neq 0$, then:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists or is $\pm\infty$.

- **Other Forms:** Convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

$$- 0 \cdot \infty \implies f(x)g(x) = \frac{f(x)}{1/g(x)}$$

$$- 1^\infty, 0^0, \infty^0 \implies y = f(x)^{g(x)} \implies \ln y = g(x) \ln f(x)$$

1.2 Maxima and Minima (One Variable)

Let f be a continuous function defined on an interval I .

- **Critical Points:** Points c in the domain where $f'(c) = 0$ or $f'(c)$ does not exist.
 - **First Derivative Test:** Let c be a critical point.
 - If $f'(x)$ changes from $+$ to $-$ at c , $f(c)$ is a local maximum.
 - If $f'(x)$ changes from $-$ to $+$ at c , $f(c)$ is a local minimum.
 - If $f'(x)$ does not change sign, c is a point of inflection.
 - **Second Derivative Test:** Let $f'(c) = 0$.
 - If $f''(c) < 0$, f has a local maximum at c .
 - If $f''(c) > 0$, f has a local minimum at c .
 - If $f''(c) = 0$, the test is inconclusive.
 - **Global Extrema on $[a, b]$:** Evaluate $f(x)$ at all critical points in (a, b) and at the endpoints $x = a, x = b$. The largest value is the absolute maximum; the smallest is the absolute minimum.
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2. Power Series (One Real Variable)

2.1 Term-wise Differentiation and Integration

Let a power series $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ have a radius of convergence $R > 0$. Then, on the interval $(a-R, a+R)$:

- **Continuity:** The function $f(x)$ is continuous, infinitely differentiable, and integrable.
- **Term-wise Differentiation:** The derivative can be found by differentiating term-by-term, and the radius of convergence remains R :

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$$

- **Term-wise Integration:** The integral can be found by integrating term-by-term, and the radius of convergence remains R :

$$\int f(x) dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}$$

- *Note:* While the radius of convergence R remains the same for the differentiated and integrated series, convergence at the exact endpoints ($x = a \pm R$) must be checked separately.
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3. Functions of Two or Three Real Variables

3.1 Total Derivative

The total derivative measures the change in a multivariable function taking into account the variations of all its variables simultaneously.

- **Total Differential:** If $z = f(x, y)$ has continuous partial derivatives, the total differential dz is:

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

For $w = f(x, y, z)$, $dw = f_x dx + f_y dy + f_z dz$.

3.2 The Chain Rule

- **Path Derivative (One independent variable):** If $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ are differentiable functions of t :

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

- **Multiple Independent Variables:** If $z = f(x, y)$ where $x = x(u, v)$ and $y = y(u, v)$:

$$\begin{aligned} \frac{\partial z}{\partial u} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ \frac{\partial z}{\partial v} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} \end{aligned}$$

- **Implicit Differentiation:** If an equation $F(x, y) = 0$ defines y implicitly as a differentiable function of x , and $F_y \neq 0$:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$